



Characterising Optical Properties of Doped Metal Complex Nanocomposite Films with PVA/PVAC for Optoelectronics

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ABSTRACT

This study investigates the optical properties of nanocomposite films made from polyvinyl alcohol (PVA), polyvinyl acetate (PVAC), and a 50/50 polyblend of PVA and PVAC, doped with varying concentrations 0, 3, 6, and 9 wt.% of the metal complex Fluoro pentaamine cobalt (III) fluoride $[\text{Co}(\text{NH}_3)_5\text{F}]\text{F}_2$ (FACF). The films were prepared using a solvent casting method and analysed with a computerised UV-VIS-IR spectrophotometer over the wavelength range of 250–850 nm. Key optical parameters, including the refractive index, extinction coefficient, and optical conductivity, were determined through absorption studies. The optical absorbance increased with higher FACF concentrations. UV-Vis spectra were used to estimate the band gap energies and investigate the electronic transitions from the valence band to the conduction band. A decrease in the band gap for allowed direct transitions was observed: from 4.0 eV to 2.55 eV for PVA, 4.1 eV to 2.6 eV for PVAC, and 4.35 eV to 2.6 eV for the polyblend as the FACF content increased. Additionally, both the extinction coefficient and refractive index increased with higher FACF concentrations. These nanocomposite films show potential for use in photovoltaic cells, optical sensors, LEDs, photodetectors and other luminescent devices, though further research is needed.

KEYWORDS

absorbance spectrum, casting method, energy gap, polyblend, polyvinyl acetate, polyvinyl alcohol

CITATION

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1. Introduction

Nowadays, polymers are widely diversified and play an important role in many aspects of human life due to their better features compared to traditional materials, such as low production costs, chemical stability, lightweight and ease of processing (Aziz *et al.*, 2017; Bhagyaraj *et al.*, 2020). Novel polymers and polymer composites have several unique qualities, including simple processing methods and optical characteristics that make them strong contenders for a variety of optoelectronic uses, including data processing, integrated optics, optical sensors, microlasers and Nanophotonics (Beckers *et al.*, 2017; Loste *et al.*, 2019). Moreover, polymers impregnated with noble metal nanoparticles exhibited unique and elevated features resulting from a distinct fusion of the intrinsic properties of the polymers and those of the metal nanoparticles (Hassan and Ah-yasari, 2019). By hydrolysing polyvinyl acetate in ethanol using potassium hydroxide, Haehnel and Hermann produced poly (vinyl alcohol) (PVA) for the first time in 1924 (Sharma *et al.*, 2023). PVA is a highly versatile material with several desirable properties, including high dielectric strength and excellent transparency and optical properties. These characteristics make it an attractive option for the preparation of electrolyte membranes, given its mechanical and chemical strength (Palanichamy *et al.*, 2023). PVA is synthetic, linear and semi-crystalline in nature. It contains a carbon chain and a hydroxyl functional group that acts as the backbone (Agrawal *et al.*, 2023). It is the most popular water-soluble polymer with the formula $(\text{C}_2\text{H}_4\text{O})_n$, and it is manufactured by saponification of poly 'vinyl ester' with sodium hydroxide (NaOH). PVA has an extremely high transmission of visible light, making it desirable in a variety of opto-electronic systems and applications and is economically and environmentally beneficial (Palanichamy *et al.*, 2023). Under acidic conditions, the OH groups in PVA tend to cross-link with aldehydes, forming hemiacetal or actual linkages chemically (Bulinski, 2021), (Zhong *et al.*, 2024).

In contrast, Poly (vinyl acetate) (PVAC) is referred to as 'white glue'. It

is a clear, glassy substance that is transparent at ambient temperature, insoluble in water, odourless, colourless, and non-toxic. PVAC is a thermos-plastic polymer. In the presence of an alkali or acid, it can be alcoholysed to produce PVA. PVAC is highly stable in the presence of light and hardly ages under the influence of light. The term 'white latex' is commonly used to describe PVAC emulsion. It can be modified directly with a variety of additives. Its mechanical strength is good; specifically, it is a water-based adhesive and does not pollute the environment (Aruldass *et al.*, 2019). Since PVAC contains a COOH group, it can be amenable to the formation of safe and biocompatible films for application in both industrial and biomedical settings (Khalifa *et al.*, 2024).

Substances called metal complexes are frequently used in analytical chemistry. They can be produced to have specific molecular geometries, making it possible to use their unique spectroscopic, physical and electrochemical properties for analytical applications (Elattar *et al.*, 2024), (Yam and Lo, 1999). A metal complex or a 'coordination complex' is a type of chemical compound that consists of a coordination centre, which is usually a basic metal atom or ion, and a surrounding group of linked neutral molecules. Alternatively, they can be anions such as fluoride or chloride, also referred to as ligands or complex agents. These ligands can produce bonds by donating electrons to the orbitals of the metal ion (Kettle, 2013). Transition metals are distinguished by the multiple oxidation states (Chan and Wong, 2013). A paramagnetic compound Fluoro penta Amine Cobalt (III) Fluoride $[\text{Co}(\text{NH}_3)_5\text{F}]\text{F}_2$ (FACF) is the metal complex utilised in this work. FACF exhibits characteristic optical properties that are primarily influenced by the ligand field around the Co^{3+} ion, leading to the formation of distinct absorption bands in the UV-visible region. The observed colour is a consequence of these electronic transitions, and spectroscopic techniques like UV-Vis and IR spectroscopy are essential for studying these properties (Van Eldik *et al.*, 1978), (Schönherr, 2004). The purpose of this study is to examine the influence of incorporating metal complexes of FACF at

varying concentrations on the optical behaviour of PVA and PVAC, both individually and in a poly blend of 50% PVA + 50% PVAC, using UV-visible radiation. The results of these measures have been subjected to investigation and discussion with regard to their potential applicability in a multitude of industrial applications.

2. Experimental Procedures

2.1. Prepared Fluoro Penta Amine Cobalt (III) Fluoride (FACF):

The preparation of FACF was conducted in accordance with the methodology described by Rabee *et al.* (2013). In a 400 ml container, (2 g) of ammonium fluoride NH_4F was dissolved totally in approximately 15 ml of pure NH_3^+ while stirring continuously. Then (3.5 g) of Cobalt fluoride (CoF_2) was added progressively to the mix. After obtaining a brown-coloured slurry, 2.5 mL of 33% Hydrogen Peroxide (H_2O_2) was gradually added. At the end of the effervescence, approximately 15 mL of concentrated hydrofluoric acid (HF) was gradually added. While stirring continuously on a hot plate, the mixture was heated to 85°C for 20 minutes, and then a Buchner funnel was used to filter out the FACF crystals after using ice to cool the mixture to room temperature.

2.2. Samples Preparation:

The polymers used in this work as the polymer matrix were poly (vinyl alcohol) (PVA), $(\text{C}_2\text{H}_4\text{O})_n$, and poly (vinyl acetate) (PVAC) $(\text{C}_4\text{H}_6\text{O}_2)_n$. The PVA used was a fine white grain with a molecular weight ranging from 30,000 to 70,000 and a purity of 99.5%. PVAC is a solid polymer granule that is white to yellow with a molecular weight of 100,000 and 99.6% pure. All of the chemicals needed to make FACF were purchased from Sigma-Aldrich. Method of Solvent casting was used to create films of PVA, PVAC and 50% PVA + 50% PVAC doped with 0, 3, 6 and 9 wt% of metal complexes of FACF. First, distilled water was added to solid PVA $(\text{C}_2\text{H}_4\text{O})_n$ to produce a PVA solution. The distilled water was used to mix the PVA emulsion for ten hours, and then stirred it with a magnetic stirrer for two hours at 70°C . Using a magnetic stirrer, the required weight fractions of FACF were distributed in distilled water for one hour and then progressively added while stirring continuously into the polymeric emulsion, leaving it under string for two hours. Subsequently, the solution was transferred to clean Petri dishes and left to gradually evaporate at room temperature for a week. Following the drying process, the thin films were peeled from the Petri dishes and kept in vacuum desiccators until utilised. Using the same steps, the remaining PVAC samples and the combination solution of 50 % PVA + 50% PVAC were prepared. The produced films had a thickness between 100 and 120 nm. The optical characteristics of the sample's preparation films were examined using UV-VIS-IR spectroscopy in the submitted work.

3. Results and Discussions

3.1. Fourier Transform Infrared Spectroscopy (FTIR):

Fourier transform infrared (FTIR) spectroscopy is a significant analytical technique for examining the structural characteristics of polymers. It offers insights into the complex interactions and associations between the diverse components present in polymeric systems. Figure 1 illustrates the FTIR spectra of PVA, PVAC and a poly blend of 50% PVA + 50% PVAC doped with 0, 3, 6 and 9 wt. % of the metal complex of FACF. Below, Tables 1, 2 and 3 simulate common expected peaks based on typical FTIR responses for PVA and PVAC polymers doped with a cobalt complex. The tables present illustrative data regarding the absorption intensities at notable wavenumbers for each sample

configuration. They comprise readings and typical peak regions observed for these materials in FTIR, encompassing prominent regions for PVA, PVAC and their doped blends.

Figure 1a. FTIR graph of PVA, PVAC and a poly blend of 50% PVA + 50% PVAC doped with 0, 3, 6 and 9 wt. % of the metal complex of FACF.

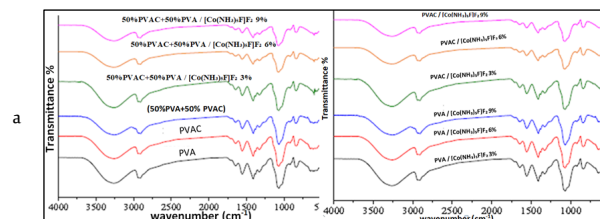


Figure 1b. FTIR graph of 50% PVA + 50% PVAC thin film.

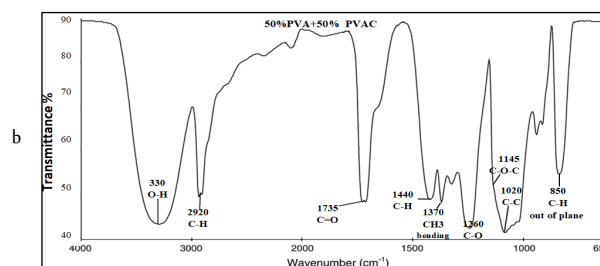


Table 1. Values of absorption peaks, peak assignments and transmission of PVA with 3, 6, 9 wt. % concentration of FACF composites; results based on FTIR analyses.

PVA			PVA / (3%) FACF			PVA / (6%) FACF			PVA / (9%) FACF		
Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)
3300	O-H stretching (hydroxyl groups)	60	3300	O-H stretching (hydroxyl groups in PVA)	57	3280	O-H stretching (hydroxyl groups in PVA)	55	3265	O-H stretching (hydroxyl groups in PVA)	53
2920	C-H stretching (asymmetrical)	65	2920	C-H stretching (asymmetrical)	63	2915	C-H stretching (asymmetrical)	60	2910	C-H stretching (asymmetrical)	57
1730	C=O stretching (carbonyl, minor)	60	1730	C=O stretching (carbonyl, minor in PVA)	58	1725	C=O stretching (carbonyl, minor in PVA)	55	1720	C=O stretching (carbonyl, minor in PVA)	53
1440	C-H bending	71	1440	C-H bending	67	1435	C-H bending	64	1430	C-H bending	61
1375	CH ₃ bending	62	1370	CH ₃ bending	59	1365	CH ₃ bending	56	1360	CH ₃ bending	54
1270	C-O stretching	50	1260	C-O stretching	52	1255	C-O stretching	50	1250	C-O stretching	47
1150	C-O-C stretching (ether linkages)	55	1140	C-O-C stretching (ether linkages)	48	1135	C-O-C stretching (ether linkages)	45	1130	C-O-C stretching (ether linkages)	42
950	C-F stretching	40	950	C-F stretching (from FACF)	43	940	C-F stretching (from FACF)	40	935	C-F stretching (from FACF)	37
			850	Co-N stretching	38	830	Co-N stretching	35	820	Co-N stretching	32
720	Out-of-plane bending vibrations	30	710	Out-of-plane bending vibrations	28	700	Out-of-plane bending vibrations	25	690	Out-of-plane bending vibrations	23

Table 2. Values of absorption peaks, peak assignments and transmission of PVAC with 3, 6, 9 wt. % concentration of FACF composites; results based on FTIR analyses.

PVAC			PVAC / (3%) FACF			PVAC / (6%) FACF			PVAC / (9%) FACF		
Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)
2920	C-H stretching (asymmetric)	70	2920	C-H stretching (asymmetric)	68	2925	C-H stretching (asymmetric)	65	2920	C-H stretching (asymmetric)	62
1735	C=O stretching (carbonyl)	65	1735	C=O stretching (carbonyl)	60	1730	C=O stretching (carbonyl)	58	1725	C=O stretching (carbonyl)	55
1440	C-H bending	70	1440	C-H bending	67	1445	C-H bending	64	1440	C-H bending	58
1375	CH ₃ bending	67	1375	CH ₃ bending	64	1370	CH ₃ bending	62	1365	CH ₃ bending	60
1260	C-O stretching	60	1260	C-O stretching	55	1265	C-O stretching	53	1260	C-O stretching	50
1150	C-O-C stretching	50	1150	C-O-C stretching	50	1150	C-O-C stretching	48	1145	C-O-C stretching	45
900	C-C stretching	45	950	C-F stretching (from FACF)	45	940	C-F stretching (from FACF)	42	900	C-F stretching (from FACF)	38
			850	Co-N stretching	40	860	Co-N stretching	38	840	Co-N stretching	35
730	Out-of-plane bending vibrations	30	720	Out-of-plane bending vibrations	30	710	Out-of-plane bending vibrations	32	700	Out-of-plane bending vibrations	28

Table 3. Values of absorption peaks, peak assignments and transmission of 50% PVA + 50% PVAC, with 3, 6, 9 wt. % concentration of FACF composites; results based on FTIR analyses

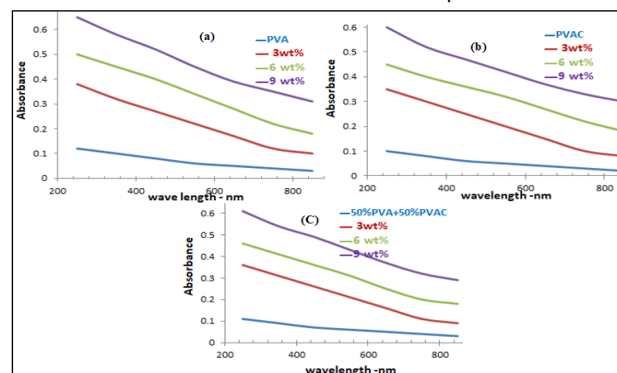
50%PVAC+50%PVA			50%PVAC+50%PVA / (3%) FACF			50%PVAC+50%PVA / (6%) FACF			50%PVAC+50%PVA / (9%) FACF		
Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)	Absorption Peak (cm ⁻¹)	Peak Assignment	Transmission (%)
3300	O-H stretching (hydroxyl groups in PVA)	53	3280	O-H stretching (hydroxyl groups in PVA)	51	3265	O-H stretching (hydroxyl groups in PVA)	49	3250	O-H stretching (hydroxyl groups in PVA)	58
29200	C-H stretching (asymmetric)	63	2915	C-H stretching (asymmetric)	61	2910	C-H stretching (asymmetric)	58	2905	C-H stretching (asymmetric)	55
1753	C=O stretching (carbonyl in PVAC)	60	1725	C=O stretching (carbonyl in PVAC)	56	1720	C=O stretching (carbonyl in PVAC)	54	1715	C=O stretching (carbonyl in PVAC)	52
1440	C-H bending (common to PVAC and PVA)	52	1440	C-H bending	49	1435	C-H bending	43	1430	C-H bending	62
1370	CH ₃ bending (PVAC)	48	1365	CH ₃ bending (PVAC)	44	1360	CH ₃ bending (PVAC)	40	1355	CH ₃ bending (PVAC)	66
1260	C-O stretching (common to PVAC and PVA)	55	1255	C-O stretching	52	1250	C-O stretching	50	1245	C-O stretching	48
1145	C-O-C stretching (ether linkages in PVA)	52	1135	C-O-C stretching (ether linkages in PVA)	50	1130	C-O-C stretching (ether linkages in PVA)	47	1125	C-O-C stretching (ether linkages in PVA)	45
1020	C-C stretching	45	935	C-F stretching (from FACF)	42	930	C-F stretching (from FACF)	39	925	C-F stretching (from FACF)	36
850	C-H out-of-plane bending	55	840	Co-N stretching	51	830	Co-N stretching	48	820	Co-N stretching	30
690	Out-of-plane bending vibrations	30	695	Out-of-plane bending vibrations	28	690	Out-of-plane bending vibrations	25	680	Out-of-plane bending vibrations	22

3.2. UV-VIS Spectra:

The spectra of absorbance for wavelengths of samples ranging from 200 to 900 nm are indicated in Figure 1 for samples of PVA, PVAC and 50 % PVA + 50% PVAC with different concentrations of 0, 3, 6, and 9 wt.% FACF. It is possible that multiple electron transitions can happen in polymers. Optical absorption measurements are used to determine the band structure, absorption edge and optical energy band gaps. Pure polymers show no signs of absorption, but by increasing the doping concentration of the FACF, the absorption gradually increases, as indicated in this figure. In other words, absorption increases with the proportion of absorbed molecules, and this rise can be explained by the minute structural distortion found in both PVA and PVAC (Abdelaziz, 2012). The FACF compound enhances the absorbance spectra of the PVA and PVAC host materials. Furthermore, as the concentration of FACF increases, the wavelength corresponding to the aforementioned shoulder-like absorption band also increases. The addition of FACF resulted in the appearance of unique peaks that could be attributed to

the formation of charge transfer complexes in the polymer due to the presence of halogens (Gadhawe and Dhawale, 2022). These peaks broaden when the concentration of FACF increases due to the high binding between the additive and the host poly. The complexation between PVA, PVAC and FACF is indicated by the red-shift behaviour observed in the composite films and could potentially result from a shift in crystal structure caused by additives (Abdelrazek and Elashmawi, 2008). Furthermore, Figure 1 illustrates how absorbance rises with the percentages of absorbed particles. From the graph, it is evident that maximum absorption is found at 9% for all samples in this study.

Figure 2. Optical absorption versus wavelength of (a) PVA, (b)PVAC and (c) 50 %PVA+50% PVAC, with 0, 3, 6, and 9 wt. % concentration of FACF composites.



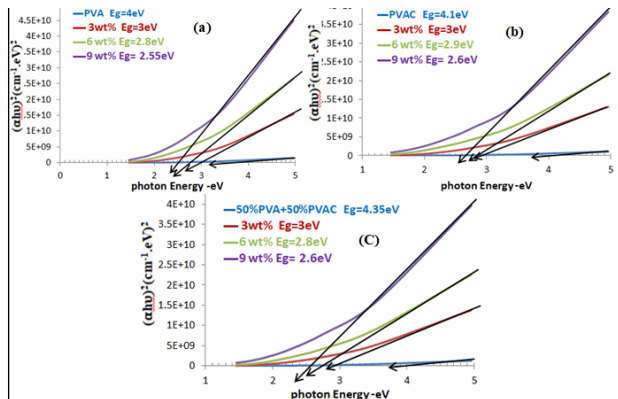
Due to the exceptional importance of the optical band gap and the energy values of the materials being studied, there have been numerous attempts to analyse and calculate the energy required for the electronic transition from the valence-band level to the conduction-band level. Based on the following experimental relationship, these models and assumptions are based on the value of the absorption coefficient α , which appears as an exponential function of the photon energy near the band edge in the semiconductor (Mohammed *et al.*, 2022):

$$\alpha h\nu = \beta (h\nu - E_{opt})^y \dots\dots\dots (1)$$

where β is a constant named the band tailing parameter, E_{op} is the energy of the optical band gap, and y is the power factor of the transition mode, which depends on the type of material being studied – whether it is direct or indirect, allowed or forbidden. The energy of the band gap is determined from the graph of $(\alpha h\nu)^{1/y}$ vs photon energy ($h\nu$). The plots of the band gap energy values obtained from Figures 3 a, b and c illustrate the allowed direct transition $y=1/2$. As seen in Figures 3 a, b and c, a reasonable linear fit is obtained for $(\alpha h\nu)^2$ vs $h\nu$. The respective values of E_{op} are obtained by extrapolating to $(\alpha h\nu)^2 = 0$ for allowed and direct transition. In the current investigation, a band gap of 4 eV was gained for pure PVA, while 4.1 eV was obtained for pure PVAC, as shown in table 4. It can be observed that across all three material systems, higher doping concentrations (in particular, the 9 wt% sample) result in a markedly elevated absorption coefficient at a given photon energy. This trend suggests that the addition of FACF enhances the material's capacity to absorb light, potentially due to an increase in electronic states that facilitate photon absorption. In each case, the absorption rate rises markedly at a particular threshold energy level. This point frequently correlates with the material's band gap energy, wherein electronic transitions from the valence to the conduction band become feasible. The figures indicate that the absorption onset occurs at approximately 2 to 3 eV, which suggests that this range may be close to the band gap energies for these doped polymers, PVA vs PVAC vs mixed (50% PVA + 50% PVAC). Each polymer matrix (PVA, PVAC and

the blend) displays distinctive absorption characteristics in the presence of the dopant. In general, the absorption coefficient for the mixed system (50% PVA + 50% PVAC) appears to increase in a more gradual manner with increasing doping concentration. In contrast, the pure PVAC and PVA systems exhibit more pronounced changes in slope with each doping level. The slope of the absorption curves undergoes a change with the level of doping, particularly at higher doping concentrations. For example, the 9 wt% doped samples exhibit a steeper curve, indicating a higher rate of increase in absorption with photon energy. This finding may be attributed to an enhanced interaction between photons and the electronic states introduced by the FACF dopant. The interaction between each polymer matrix and the dopant is unique, influencing the absorption behaviour in a manner that can be tailored by modifying the dopant concentration and polymer composition. The observed reduction in the optical energy gap with increasing doping ratio can be attributed primarily to the introduction of impurity-induced states (band tails), enhanced carrier interactions and disorder effects that modify the band structure. At lower doping levels, the optical energy gap may initially appear constant or even increase slightly due to the Burstein-Moss shift. However, as doping intensifies, the cumulative effects of band gap renormalisation, carrier-phonon interactions and localised states in the band tails result in a narrowed optical energy gap. This behaviour is highly relevant in applications such as solar cells and LEDs, where tuning the band gap by doping can improve device performance and efficiency (Zidan *et al.*, 2003). Table 4 presents the obtained values of the energy band gaps.

Figure 3. Energy gaps of allowed direct transition $(\alpha h\nu)^2$ against photon energy for PVA, PVAC and 50% PVA+50% PVAC, with 0, 3, 6, and 9 wt. % concentration of FACF composites

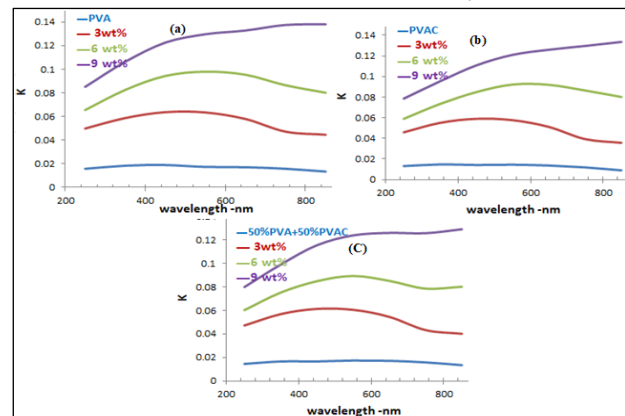


The variation in the coefficient of excitation k with wavelength provides confirmation of the interaction between the energy of the medium and the photon. The value of K was determined by applying the following formula (Balkanski, 1980):

$$K = \alpha\lambda / 4\pi \dots \dots (2)$$

The dependence of K on the wavelength 190–800 nm of pure for both PVA, PVAC and 50% PVA+50% PVAC / 0, 3, 6, 9 wt% of FACF compound samples is illustrated in Figure 4 a, b and c. The graph shows a decline in the coefficient of excitation values for the pure PVA, PVAC and 50% PVA+50% PVAC samples across 290–800 nm wavelengths, while there is an increase in the coefficient of excitation values for the 3, 6, and 9 wt%. As the concentration of FACF increased, the coefficient of excitation also increased due to the higher concentration of the additive, resulting in a greater coefficient of absorbance.

Figure 4. Extinction coefficient versus wavelength for (a) PVA, (b) PVAC, and (c) 50% PVA+50% PVAC, with 0, 3, 6, and 9 wt. % concentration of FACF composites.



The refractive index (n) is an essential optical characteristic of every polymer. It is closely linked to a material's optical, magnetic and electrical characteristics, making it valuable for researching the chemical, physical and molecular features of a polymer using optical methods. The index of refraction is calculated according to the following formula (Salman *et al.*, 2015):

$$n = \sqrt{\frac{4R - K^2}{(R - i)^2}} - \frac{(R + 1)}{(R - 1)} \dots \dots (3)$$

where R is reflectance. The plans are shown in Figures 5 a, b and c, which represent the variations of the refraction index of PVA, PVAC and 50% PVA+50% PVAC, with different doping concentrations of 0, 3, 6, and 9 wt.% FACF thin films, respectively, in the investigated range of wavelengths at room temperature. In Figure 5, it is evident that for all prepared samples, the index of refraction decreases as the wavelength increases. Conversely, the figure demonstrates that the index of refraction (n) increases with the addition of the additive percentage FACF, which is in line with the growth in packing density due to the rise in FACF content (Rabee *et al.*, 2013). In other words, this result can be attributed to the occurrence of additional atomic refractions as a consequence of the growth linear polarisation, which is consistent with the Lorentz formula (Ahmad *et al.*, 2007).

Figure 5. The refraction of the index vs wavelength for (a) PVA, (b) PVAC, and (c) 50% PVA+50% PVAC, with 0, 3, 6, and 9 wt.% concentration of FACF composites.

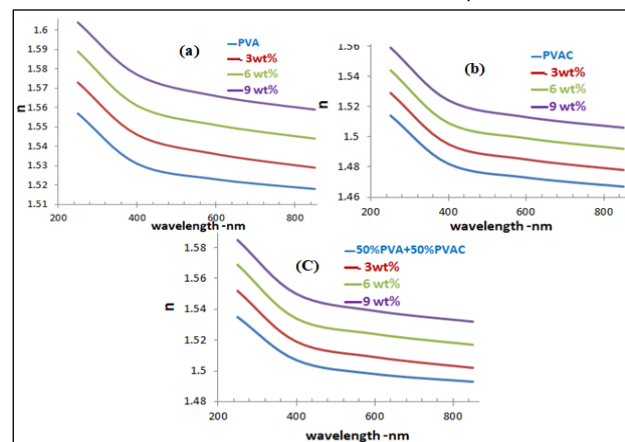


Figure 6 shows the deviation of optical conductivity σ as a function of photon energy for PVA, PVAC and 50% PVA+ 50% PVAC, with different doped concentrations of 0, 3, 6, and 9 wt.% FACF thin films. It can be observed that pure PVA and PVAC have nearly constant

conductivity σ up to about 5.6 eV of photon energy, then it rises as the photon energy increases. The higher absorption behavior of the polymer composite films, resulting from the increased concentration of FACF, leads to an enhancement in optical conductivity. The band gap energy decreases while optical conductance increases due to an increase in the FACF concentration caused by a growth in the total of mobile charge carriers and an increase in the amorphous nature of the host polymer.

Figure 6. Optical conductivity σ as a function of photon energy for (a) PVA, (b) PVAC and (c) 50%PVA+50%PVAC, with 0, 3, 6, and 9 wt.% concentration of FACF composites.

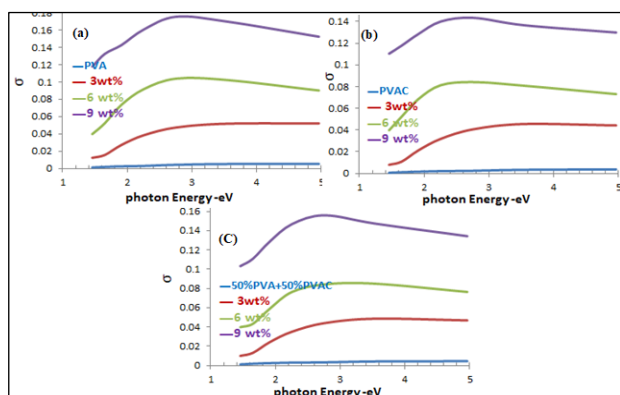


Table 4. Energy gap values of the allowed direct transition of PVA, PVAC and 50% PVA+50% PVAC, with 0, 3, 6, and 9 wt.% concentration of FACF composites.

Additive concentrate %	PVA/[Co(NH ₃) ₅ F]F ₂ E _g (ev) of allowed direct transitions	PVAC/[Co(NH ₃) ₅ F]F ₂ E _g (ev) of allowed direct transitions	50%PVA+50%PVAC/[Co(NH ₃) ₅ F]F ₂ E _g (ev) of allowed direct transitions
0	4	4.1	4.35
3	3	3	3
6	2.8	2.9	2.8
9	2.55	2.6	2.6

4. Conclusions

Polymer films based on PVA, PVAC and 50% PVA+50% PVAC, with different concentrations; 0, 3, 6, and 9 wt.% FACF, were prepared using a solvent casting procedure. The optical characteristics were analysed depending on the additive concentrations. The UV outcomes demonstrated that FACF is an effective additive for improving the optical characteristics of both PVA and PVAC, whether used individually or in combination. The absorbance of the PVA and PVAC increased as the weight percentages of FACF increased. The optical properties also show that as the concentration of FACF increases, the band gap energy of the host polymer matrix decreases. This is thought to be due to a higher level of disorder as a result of the increase in the local state within the optical band gap for both PVA and PVAC films. The appearance of the overlap corresponds to the increase in doping concentrations for all samples, which is caused by additional complexation with the polymer matrix. These overlaps confirm that the energy band gap decreases with increasing FACF content in the polymer matrix. The optimal optical characteristics were achieved at 9 wt.% FACF for all prepared samples. The impact of varying concentrations of FACF on the optical performance of nanocomposite films has been studied, with a particular focus on the coefficients of extinction, optical conductivity, and index of refraction. Finally, the field of metal-doped PVA research is still open. Owing to its significance in novel high-tech uses, its effectiveness in optoelectronics has been a significant focus of scientific investigation.

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Abd Al-Kader, an Iraqi, holds a PhD in physics, specialising in thin films, from Anbar University, College of Science. His research, published in Scopus-indexed journals, focuses on light-matter interactions in various materials, particularly in optics and plasmas. He is currently a physics teacher at the Anbar Education Directorate and has experience as a research evaluator for several academic journals. He is also deeply interested in green synthesis and nanomaterials, exploring sustainable methods for material development.

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